Performance Testing

Date	29 june 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Local host
Type of Testing	Unit Testing, Performance Testing, model testing
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	local
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Launch the OptiCrop application	1	1 min	Home page loads successfully.
2	Enter valid soil and environmental data	1	1 min	Recommended crop is predicted accurately.
3	Verify the Predict button functionality	1	1 min	Prediction process completes successfully.
4				

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	2 sec	Pass	NO
2	Response Time (Max)	< 5 seconds	2 sec	pass	NO
3	Throughput (Req/sec)		2 sec	Pass	NO
4	Error Rate	< 1%	1%	pass	NO
5	CPU Utilization	< 80%	<80%	pass	NO
6	Memory Utilization	< 80%	<80%	pass	NO

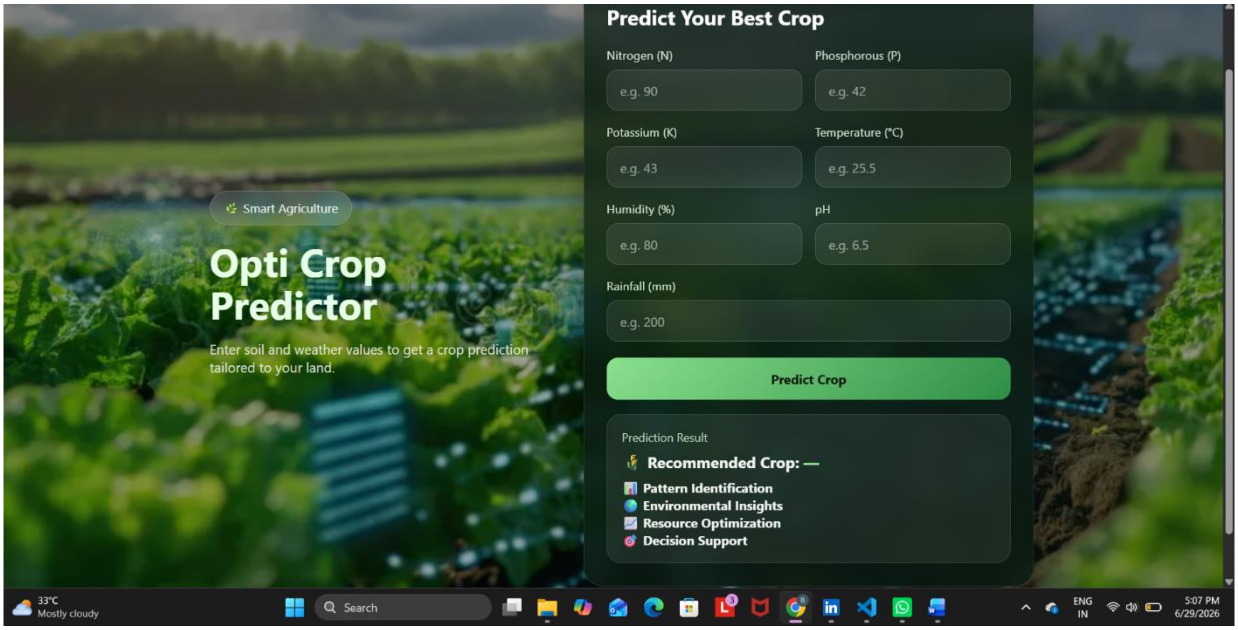
Step 4: Observations & Analysis

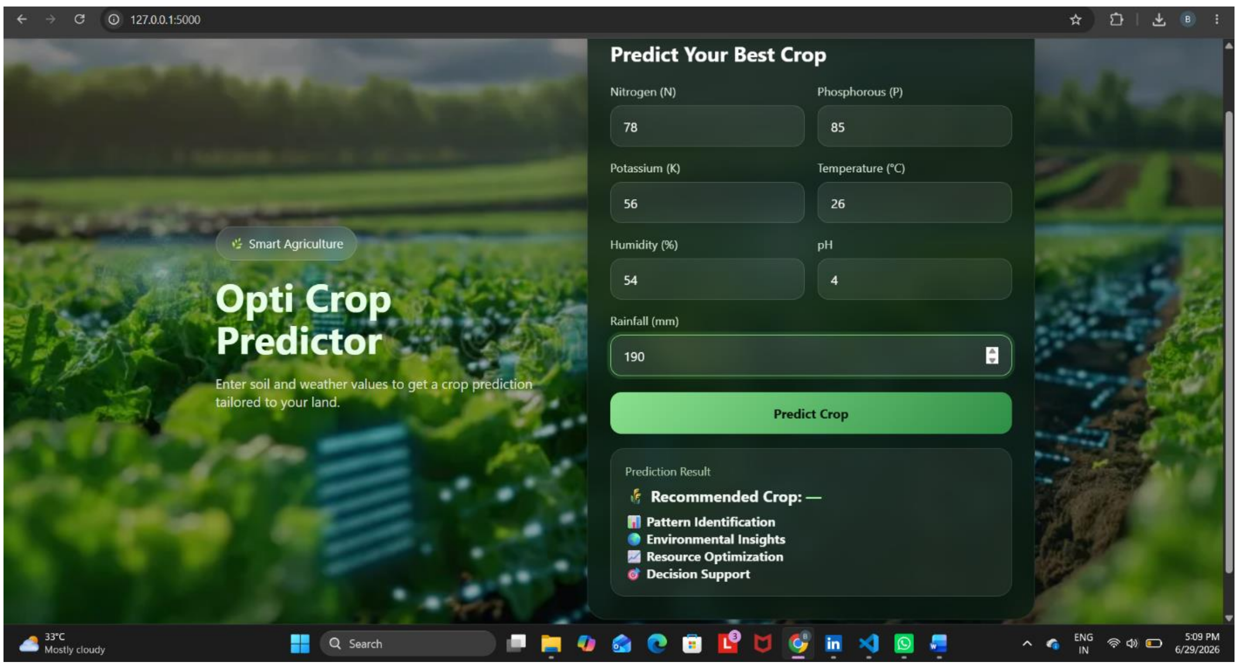
The application loads successfully without errors.

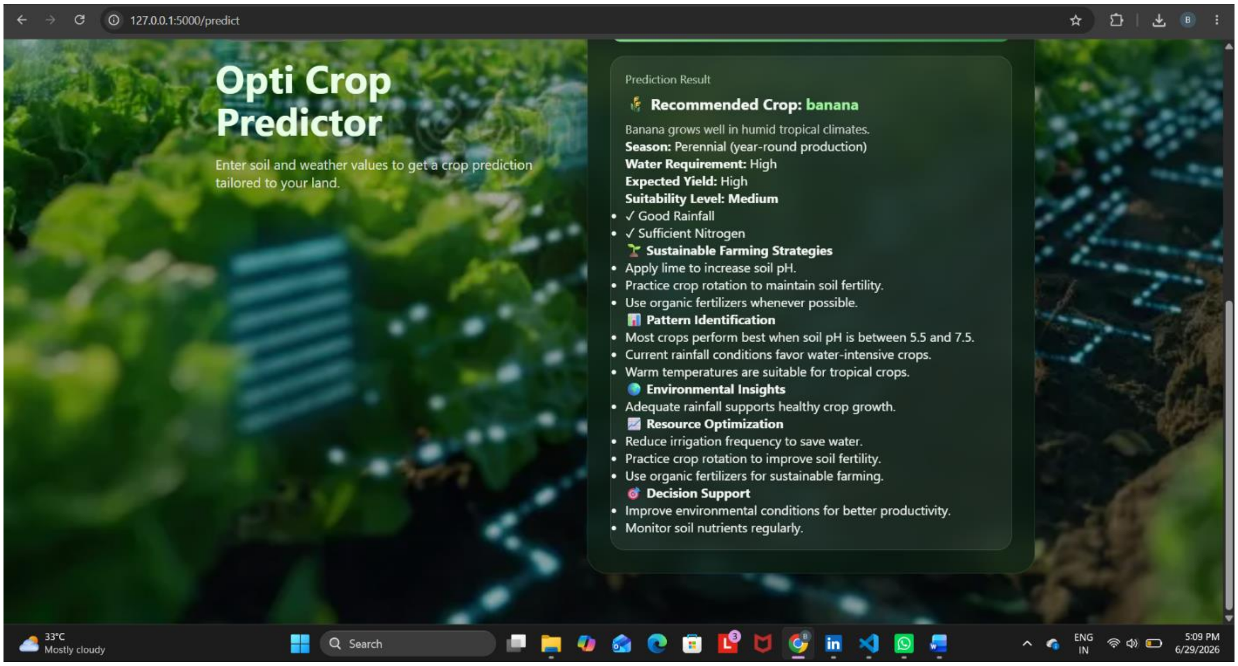
The Flask backend and frontend integration are functioning correctly.

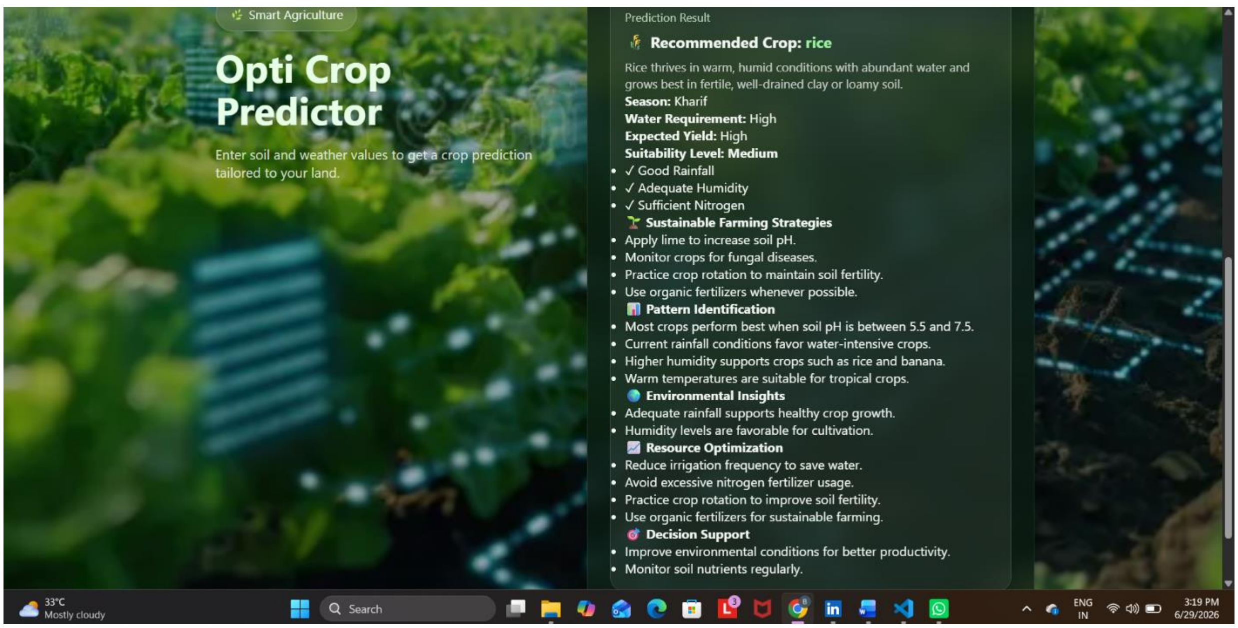
Prediction results are displayed quickly.

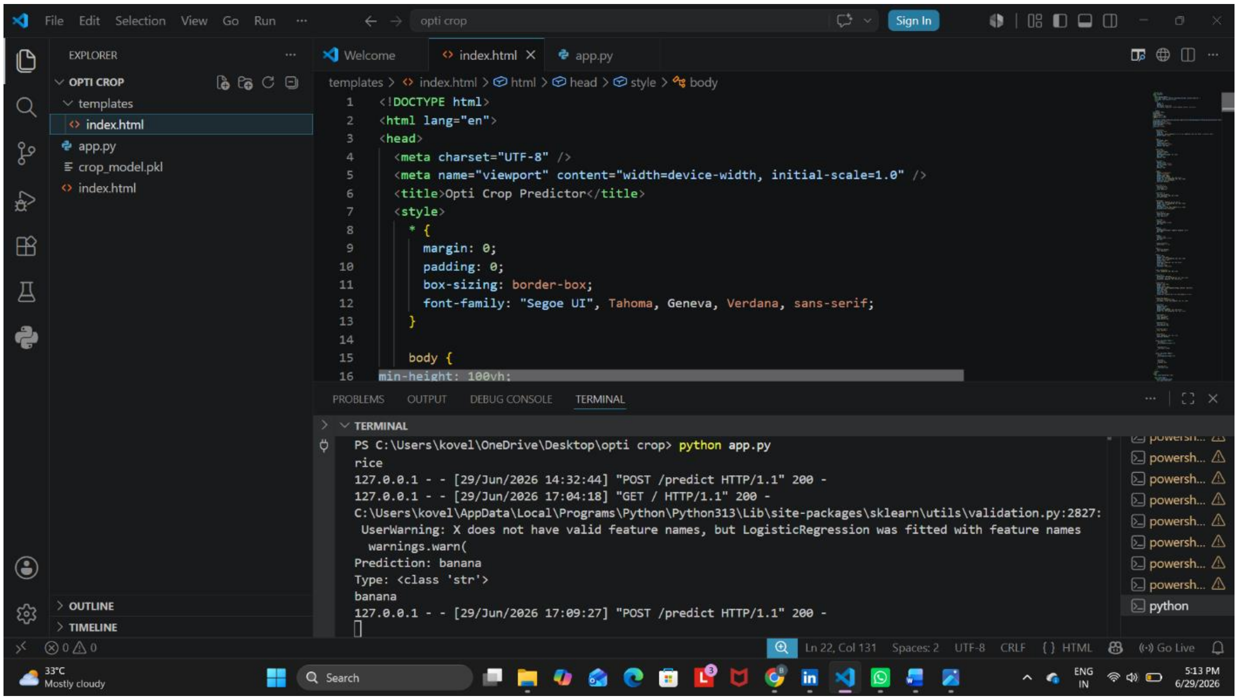
Step 5: Screenshots / Evidence


Predict Your Best Crop
Nitrogen (N)
e.g. 90
Phosphorous (P)
e.g. 42
Potassium (K)
e.g. 43
Temperature (°C)
e.g. 25.5
Humidity (%)
e.g. 80
pH
e.g. 6.5
Rainfall (mm)
e.g. 200
Predict Crop
Prediction Result
Recommended Crop: —
Pattern Identification
Environmental Insights
Resource Optimization
Decision Support
Smart Agriculture
Opti Crop Predictor
Enter soil and weather values to get a crop prediction tailored to your land.
33°C
Mostly cloudy
Search
ENG
IN
5:07 PM
6/29/2026


127.0.0.1:5000
Predict Your Best Crop
Nitrogen (N)
78
Phosphorous (P)
85
Potassium (K)
56
Temperature (°C)
26
Humidity (%)
54
pH
4
Rainfall (mm)
190
Predict Crop
Prediction Result
Recommended Crop: —
Pattern Identification
Environmental Insights
Resource Optimization
Decision Support
Smart Agriculture
Opti Crop Predictor
Enter soil and weather values to get a crop prediction tailored to your land.
33°C
Mostly cloudy
Search
ENG
IN
5:09 PM
6/29/2026


127.0.0.1:5000/predict
Opti Crop Predictor
Enter soil and weather values to get a crop prediction tailored to your land.
Prediction Result
Recommended Crop: banana
Banana grows well in humid tropical climates.
Season: Perennial (year-round production)
Water Requirement: High
Expected Yield: High
Suitability Level: Medium
✓ Good Rainfall
✓ Sufficient Nitrogen
Sustainable Farming Strategies
Apply lime to increase soil pH.
Practice crop rotation to maintain soil fertility.
Use organic fertilizers whenever possible.
Pattern Identification
Most crops perform best when soil pH is between 5.5 and 7.5.
Current rainfall conditions favor water-intensive crops.
Warm temperatures are suitable for tropical crops.
Environmental Insights
Adequate rainfall supports healthy crop growth.
Resource Optimization
Reduce irrigation frequency to save water.
Practice crop rotation to improve soil fertility.
Use organic fertilizers for sustainable farming.
Decision Support
Improve environmental conditions for better productivity.
Monitor soil nutrients regularly.
33°C
Mostly cloudy
Search
ENG
IN
5:09 PM
6/29/2026


Smart Agriculture
Opti Crop Predictor
Enter soil and weather values to get a crop prediction tailored to your land.
Prediction Result
Recommended Crop: rice
Rice thrives in warm, humid conditions with abundant water and grows best in fertile, well-drained clay or loamy soil.
Season: Kharif
Water Requirement: High
Expected Yield: High
Suitability Level: Medium
✓ Good Rainfall
✓ Adequate Humidity
✓ Sufficient Nitrogen
Sustainable Farming Strategies
Apply lime to increase soil pH.
Monitor crops for fungal diseases.
Practice crop rotation to maintain soil fertility.
Use organic fertilizers whenever possible.
Pattern Identification
Most crops perform best when soil pH is between 5.5 and 7.5.
Current rainfall conditions favor water-intensive crops.
Higher humidity supports crops such as rice and banana.
Warm temperatures are suitable for tropical crops.
Environmental Insights
Adequate rainfall supports healthy crop growth.
Humidity levels are favorable for cultivation.
Resource Optimization
Reduce irrigation frequency to save water.
Avoid excessive nitrogen fertilizer usage.
Practice crop rotation to improve soil fertility.
Use organic fertilizers for sustainable farming.
Decision Support
Improve environmental conditions for better productivity.
Monitor soil nutrients regularly.
33°C
Mostly cloudy
Search
ENG
IN
3:19 PM
6/29/2026



File Edit Selection View Go Run
opti crop
Sign In
EXPLORER
OPTI CROP
templates
index.html
app.py
crop_model.pkl
index.html
Welcome
index.html
app.py
templates > index.html > html > head > style > body
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Opti Crop Predictor</title>
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
}
body {
min-height: 100vh;
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
TERMINAL
PS C:\Users\kovel\OneDrive\Desktop\opti crop> python app.py
rice
127.0.0.1 - - [29/Jun/2026 14:32:44] "POST /predict HTTP/1.1" 200 -
127.0.0.1 - - [29/Jun/2026 17:04:18] "GET / HTTP/1.1" 200 -
C:\Users\kovel\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py:2827:
UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
warnings.warn(
Prediction: banana
Type: <class 'str'>
banana
127.0.0.1 - - [29/Jun/2026 17:09:27] "POST /predict HTTP/1.1" 200 -
powersh...
python
OUTLINE
TIMELINE
Ln 22, Col 131 Spaces: 2 UTF-8 CRLF { } HTML Go Live
33°C
Mostly cloudy
Search
ENG
IN
5:13 PM
6/29/2026